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Audit of the eta/Frobenius lemma (Theorem C) of the *Signed Weighted Residue Note*

Audit package v1.0 — 3 November 2026. AI-assisted referee-style audit (Claude, Anthropic), prepared at the request of D. Jordan. Not a substitute for specialist refereeing. The inherited package `Signed_Weighted_Residue_v1_0` was read but **not modified**; this audit lives in a separate tree.

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0. Verdict

Correct after specified repairs. No mathematical error was found in the lemma or in the proof strategy. Every step either (a) is proved here in full from fixed conventions, or (b) is a precisely cited standard import. The repairs are foundational/expository, not corrections of a wrong claim:

#	Repair	Why it was needed
R1	Replace the Gordon–Hughes–Newman citation by a complete derivation of the character of $G = \eta(24\ell\tau)/\eta(24\tau)$ from the Petersson/Knopp closed form of the eta multiplier (§2).	The sign $\left(\frac{\ell}{p}\right)$ is the entire content of the lemma; it should not rest on a cited character formula whose conventions ($d < 0$, Kronecker vs Jacobi, $\ell = 2$) are not spelled out.
R2	Fix the diamond-operator and Frobenius conventions once and for all: $\langle d\rangle(E,\iota) = (E, \iota\circ[d])$, $G\circ\langle d\rangle = \chi_\ell(d)G$ (no inverse), $\sigma(\tilde{x}) = \langle -p^{-1}\rangle\tilde{x}$ (§4).	The note wrote $d^{\pm1}$ and $-p^{\mp1}$; harmless here because χ_ℓ is even and quadratic, but a proof should not contain unresolved signs.

#	Repair	Why it was needed
R3	Replace the descent/integrality sentence of Fact I5 by: (i) a two-line q-expansion principle at the $\mathbb{Z}[1/M]$-rational cusp of the μ_M-model giving $G \in \mathbb{Q}(X_1(M))$; (ii) the moduli description $D_\ell(E,C) = \ell^{12}\Delta(E,\pi^*\omega')/\Delta(E',\omega')$ as a unit on $Y_0(\ell)_{\mathbb{Z}[1/\ell]}$; (iii) the moduli definition of the degeneracy map δ with $G^{24} = \delta^* D_\ell^{-1}$; (iv) normality of $Y_1(M)_{\mathbb{Z}_{(p)}}$ plus the monic equation $T^{24} - \delta^* D_\ell^{-1}$ to get $G \in \mathcal{O}(Y_1(M)_{\mathbb{Z}_{(p)}})^\times$ (§3).	Fact I5 said "a function on this model defined over $\mathbb{Z}_{(p)}$" without saying where it is regular, and invoked unit leading coefficients at *all* cusps, which were never computed. The monic-equation route (suggested in the brief) needs only the cusp ∞.
R4	Replace "E_1 is 24-isogenous to E, so it has a Deuring model" by "E_1 is supersingular, so $j(E_1)$ has a model with $\pi = [-p]$", with a proof that $\pi_{p^2} = p \cdot u$, $u \in \mathrm{Aut}(E)$, for every supersingular model over $\mathbb{F}_{p^2}$ (§4).	The 24-isogeny is irrelevant; what matters is that changing the model of E_1 does not change the geometric point $\tilde{x}$ or the value $G(\tilde{x})$, and that ι need not be $\mathbb{F}_{p^2}$-rational.
R5	State explicitly which object "$\Delta(\tau)/\Delta(\ell\tau)$ evaluated at x" is (eq. (7.2) of the replacement section), so the claim is about the weight-0 moduli function and not about discriminants of independently chosen models.	Requested by the brief; the note's Lemma `Bisog` already implies it but the lemma statement did not say so.

The replacement LaTeX section (`replacement_section/eta_section_replacement.tex`) incorporates R1–R5 and compiles against the note's preamble (`standalone_wrapper.pdf`).

Priority. The exact signed statement $D_\ell^{(p^2-1)/24} = \left(\frac{\ell}{p}\right)$ at all geometric supersingular points was **not found** in any inspected source. Ono (2026, v2, 3 July 2026) proves the $(p^2-1)/12$ statement (his Thm 1.2), proves $\Delta(E,\omega)^{(p^2-1)/12} = (-1)^{(p+1)/2}$ on Deuring models (Thm 1.3), and in Remark 4 explicitly observes that the $(p^2-1)/24$ power can be -1 without giving a formula. The $(p^2-1)/12$ statement for $\mathbb{F}_{p^2}$-isogenous curves is older: Baker (1998, §2), from the Gross–Landweber congruence. The isogeny relation $B(E') = \ell\, B(E)$ (the note's Lemma `Bisog`) is Robert (1980, Théorème B/Lemme 7) as quoted in Baker (1998, Lemma 2.2). The constancy of $c(E) = E_{p+1}^{(p-1)/2}/\Delta^{(p^2-1)/24}$ was not found either. **An unsuccessful search does not establish novelty**; Robert (1980) could not be read in full (see §6).

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1. The exact target

For primes $p \geq 5$, $\ell \neq p$, $k = (p^2-1)/24$, and $x = (E,C)$ a geometric supersingular point of $X_0(\ell)$,

$$D_\ell(x)^k = \left(\frac{\ell}{p}\right), \qquad D_\ell(E,C) := \ell^{12}\,\frac{\Delta(E,\pi^*\omega')}{\Delta(E',\omega')}, \quad \pi : E \to E' = E/C.$$

D_ℓ is independent of ω' (both discriminants have weight 12), is an invertible function on $Y_0(\ell)$ over $\mathbb{Z}[1/\ell]$, and its q-expansion at the cusp $(\mathrm{Tate}(q), \mu_\ell)$ is $\Delta(q)/\Delta(q^\ell)$ (Tate curve: $\pi^*(dt'/t') = \ell\, dt/t$). So D_ℓ *is* the weight-0 modular function $\Delta(\tau)/\Delta(\ell\tau)$ in the only sense in which it can be evaluated at a point of $X_0(\ell)_{\mathbb{F}_p}$. Since $\ell^{12k} = \ell^{(p^2-1)/2} = 1$ in $\mathbb{F}_p$, the target is equivalent to $(\Delta(E,\pi^*\omega')/\Delta(E',\omega'))^k = \left(\frac{\ell}{p}\right)$, i.e. **the Vélu-normalised discriminant ratio is a 24th power in $\mathbb{F}_{p^2}^\times$ iff $\left(\frac{\ell}{p}\right) = 1$.** The audit of Theorem C below is independent of the weighted-residue theorem (no residue, Hecke polynomial or Kaneko–Zagier input is used).

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2. The analytic multiplier calculation

2.1 The eta-quotient theorem and its hypotheses

The note invoked Gordon–Hughes–Newman in the form of Ono (2004, Thm 1.64): for $f = \prod_{\delta|N} \eta(\delta\tau)^{r_\delta}$ with $k = \frac{1}{2}\sum r_\delta \in \mathbb{Z}$, $\sum \delta r_\delta \equiv 0$ and $\sum (N/\delta) r_\delta \equiv 0 \pmod{24}$, one has $f(\gamma\tau) = \chi(d)(c\tau+d)^k f(\tau)$ on $\Gamma_0(N)$ with $\chi(d) = \left(\frac{(-1)^k s}{d}\right)$, $s = \prod \delta^{r_\delta}$. Negative r_δ are allowed; $k = 0$ is allowed. For G: $N = 576\ell$, $\delta \in \{24, 24\ell\}$, $r_{24} = -1$, $r_{24\ell} = 1$: $k = 0$; $\sum \delta r_\delta = 24(\ell - 1)$; $\sum (N/\delta) r_\delta = 24 - 24\ell$; $s = \ell$. All hypotheses hold, including for $\ell = 2, 3$ (where $s \mid 24$). The level 576ℓ is necessary for the second congruence: with $N = 24\ell$ one gets $1 - \ell \not\equiv 0$. **The citation is correct as used.**

2.2 Complete derivation (replacement for the citation)

Petersson/Knopp closed form (Knopp, 1970, Ch. 4 §4.1; Dedekind-sum form in Apostol, 1990, Thm 3.4): for $c > 0$, $\eta(\gamma\tau) = \nu(\gamma)(c\tau+d)^{1/2}\eta(\tau)$ with

$$\nu(\gamma) = \begin{cases} \left(\frac{d}{c}\right) e\left(\frac{(a+d)c - bd(c^2-1) - 3c}{24}\right) & c \text{ odd} \\ \left(\frac{c}{d}\right) e\left(\frac{(a+d)c - bd(c^2-1) + 3d - 3 - 3cd}{24}\right) & c \text{ even} \end{cases}$$

(Kronecker symbols). *Verification:* this closed form agrees with the exact Dedekind-sum multiplier on 400 random matrices, both parities of c and both signs of d (they differ by the fixed branch constant $e(1/8)$ from $(-i)^{1/2}$), and the Dedekind-sum multiplier agrees with 50-digit `mpmath` numerics to 3×10^{-50} (`results/eta_multiplier_audit.json`).

For $\gamma \in \Gamma_0(576\ell)$, $c > 0$: $24\gamma\tau = \gamma_1(24\tau)$ and $24\ell\gamma\tau = \gamma_2(24\ell\tau)$ with $\gamma_1 = \begin{pmatrix} a & 24b \\ c/24 & d \end{pmatrix}$, $\gamma_2 = \begin{pmatrix} a & 24\ell b \\ c/(24\ell) & d \end{pmatrix}$, identical automorphy factors $(c\tau+d)^{1/2}$. Both lower-left entries are divisible by 24 (even), so the even-c line applies to both. In $\nu(\gamma_2)/\nu(\gamma_1)$ every term of the exponent is a multiple of 24 ($24 \mid c_i$ kills $(a+d)(c_2 - c_1)$ and $3d(c_2 - c_1)$; $24 \mid b_i$ kills the middle term), so the exponentials cancel, and the symbols give $\left(\frac{c_2}{d}\right)\left(\frac{c_1}{d}\right) = \left(\frac{\ell c_2^2}{d}\right) = \left(\frac{\ell}{d}\right)$ since $c_1 = \ell c_2$, $\gcd(c_2, d) = 1$. The cases $c = 0$ and $c < 0$ reduce to this (G has an integral q-expansion; $-\gamma$ acts as γ; $\left(\frac{\ell}{-1}\right) = 1$). **Hence** $G(\gamma\tau) = \left(\frac{\ell}{d}\right) G(\tau)$ **on** $\Gamma_0(576\ell)$, with the Kronecker symbol, for every prime ℓ including 2 and 3.

2.3 The character

$\chi_\ell(d) = \left(\frac{\ell}{d}\right)$ on $(\mathbb{Z}/576\ell)^\times$ is, by quadratic reciprocity, the Kronecker character of $\mathbb{Q}(\sqrt{\ell})$: conductor ℓ ($\ell \equiv 1 \bmod 4$), 4ℓ ($\ell \equiv 3 \bmod 4$), 8 ($\ell = 2$), all dividing $576\ell = 2^6 \cdot 3^2 \cdot \ell$. It is even (real quadratic field; directly $\left(\frac{\ell}{-1}\right) = 1$; and necessarily so since $-I$ acts trivially), quadratic, and $\chi_\ell(p) = \left(\frac{\ell}{p}\right)$ is the Legendre symbol for the prime $p \nmid M$. Verified exhaustively for all $d \bmod 576\ell$, $\ell \le 47$ (`eta_multiplier_audit.json: kronecker_character_identification`).

2.4 The change of variables 24τ

The note's Lemma `mult` is stated for $y = \eta(\tau)/\eta(\ell\tau)$ and γ with $24\ell \mid c$, $24 \mid b$: $y(\gamma\tau) = \left(\frac{\ell}{d}\right) y(\tau)$. Since $y(\tau) = 1/G(\tau/24)$ and $\gamma\tau/24 = \tilde{\gamma}(\tau/24)$ with $\tilde{\gamma} = \begin{pmatrix} a & b/24 \\ 24c & d \end{pmatrix} \in \Gamma_0(576\ell)$, this is equivalent to §2.2 (using $\chi_\ell^{-1} = \chi_\ell$). Verified on 20 random matrices per $\ell \le 47$. **Negative control:** with $24\ell \mid c$ but $24 \nmid b$ the multiplier of y is *not* $\left(\frac{\ell}{d}\right)$ in 24–29 of 30 random cases per level, so the hypothesis $24 \mid b$ (equivalently the level 576ℓ rather than 24ℓ) is needed.

2.5 Diamond convention

Fixed in §3.2 below: $x_{\gamma\tau} = \langle d \rangle x_\tau$ with $\langle d \rangle (E, \iota) = (E, \iota \circ [d])$, so $G \circ \langle d \rangle = \chi_\ell(d) G$ with **no inverse**. (With the convention $\langle d \rangle (E, \iota) = (E, \iota \circ [d^{-1}])$ one would get $\chi_\ell(d^{-1})$ and $\sigma = \langle -p \rangle$; the product is the same. The replacement section uses one convention throughout.)

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3. Algebraicity, descent and reduction

3.1 Which model

The μ_M-model: $Y_1(M)$, $M = 576\ell \geq 4$, the fine moduli scheme over $\mathbb{Z}[1/M]$ of pairs $(E, \iota : \mu_M \hookrightarrow E)$, with smooth proper compactification $X_1(M)$ (Deligne & Rapoport, 1973, IV–V; Katz, 1976, §1 "$\Gamma_{00}(N)^{\text{arith}}$"; Gross, 1990, §2, Prop. 2.1 — *verified from the text*: "pairs $(E, \alpha) \ldots \alpha : \mu_N \to E_N$ an embedding of group schemes"; the Tate curve with Id : $\mu_N \to \mathbb{G}_m/q^{\mathbb{Z}}$ gives a $\mathbb{Z}[1/N]$-section ∞ with formal parameter q). Analytically $\tau \mapsto (E_\tau, \iota_\tau)$, $\iota_\tau(\zeta_M) = 1/M$; under $E_\tau \cong \mathbb{C}^\times/q^{\mathbb{Z}}$ this is the canonical embedding, so Fourier expansions are q-expansions at ∞.

Relation to the point model $X_1(M)'$ (pairs (E, P)): isomorphic over $\mathbb{Z}[1/M]$ via $(E, P) \mapsto (E/\langle P\rangle,\ \mu_M \cong E[M]/\langle P\rangle)$ (Weil pairing); Gross (1990, §7, p. 470) also records that X and X' become isomorphic over $\mathbb{Z}[1/N][\zeta_N]$ by the identity-type identification. Neither isomorphism commutes with the uniformisations by $\mathfrak{H}$ in the way one might hope, which is why rationality statements cannot be transferred verbally.

3.2 Field of definition of G and the q-expansion principle

μ_M-model. G is holomorphic, non-vanishing on $\mathfrak{H}$, $\Gamma_1(M)$-invariant (§2.2 with $d \equiv 1$), meromorphic at cusps, with q-expansion $q^{\ell-1}\prod(1-q^{24\ell n})/(1-q^{24n}) \in q^{\ell-1}\mathbb{Z}[[q]]$. Descent theorem used: *the cusp ∞ and its formal parameter q are $\mathbb{Q}$-rational*; hence for $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, $G^\sigma - G$ has zero expansion at ∞ and vanishes. So $G \in \mathbb{Q}(X_1(M))$. (Replacement section, Lemma 7.4.)

Point model — the note's assertion verified, not assumed. In $X_1(M)'$ the cusp $(\mathrm{Tate}(q), \zeta_M)$ is only a $\mathbb{Z}[1/M, \zeta_M]$-section (Conrad, n.d., Ex. 2.3: "for the cusp ∞ we have only obtained it as a section over $\mathbb{Z}[1/N, \zeta_N]$"; Loeffler, n.d., §3.5: "$Y_1(N)$ is not characterised over $\mathbb{Q}$ by having q-expansions" in $\mathbb{Q}((q))$). σ_a moves ∞ to $(\mathrm{Tate}(q), \zeta_M^a) = \langle a\rangle'\infty$, so the Galois argument gives $G^{\sigma_a} = \chi_\ell(a)^{-1}G$: G is defined over the fixed field of $\ker\chi_\ell$, i.e. over $\mathbb{Q}(\sqrt{\ell})$, and its reduction over $\mathbb{F}_p(\sqrt{\ell}) \subseteq \mathbb{F}_{p^2}$. **The note's parenthetical claim is correct.** It is also unnecessary once the μ_M-model is used, and in either model the p^2-Frobenius fixes the field of definition of $\overline{G}$, so the Frobenius computation goes through verbatim.

3.3 Extension as an invertible function over $\mathbb{Z}_{(p)}$ (the repair suggested in the brief)

1. D_ℓ is a unit on $Y_0(\ell)_{\mathbb{Z}[1/\ell]}$ by its moduli definition (§1): $\Delta(E, \omega)$ is a unit on any base for any elliptic curve.
2. $\delta : (E, \iota) \mapsto (E/\iota(\mu_{24}),\ \iota(\mu_{24\ell})/\iota(\mu_{24}))$ is a $\mathbb{Z}[1/M]$-morphism $Y_1(M) \to Y_0(\ell)$; on $\mathbb{C}$-points it is $\tau \mapsto 24\tau$ (multiplication by 24 maps $E_\tau \to E_{24\tau}$ with kernel $\langle 1/24\rangle = \iota_\tau(\mu_{24})$ and carries $\langle 1/(24\ell)\rangle$ onto $\langle 1/\ell\rangle$). Hence $G^{24} = \delta^* D_\ell^{-1}$ in $\mathbb{Q}(X_1(M))$ (equal on $\mathbb{C}$-points). **Verified moduli-theoretically; surjectivity on $\overline{\mathbb{F}}_p$-points is constructed explicitly** (replacement Lemma 7.6): given (E, C), take $E_1 = E/K$ for a cyclic K of order 24 and a cyclic $Z \subset E_1$ of order M with $Z[24] = E[24]/K$ and image ℓ-part C.
3. $R = \mathcal{O}(Y_1(M)_{\mathbb{Z}_{(p)}})$ is a normal domain (smooth over $\mathbb{Z}_{(p)}$; flat with integral generic fibre). $G \in \mathrm{Frac}(R)$ is a root of the monic $T^{24} - F$, $F = \delta^* D_\ell^{-1} \in R^\times$. So $G \in R$ and $G^{-1} = G^{23}F^{-1} \in R$.

What this proves: $\overline{G} \in \mathcal{O}(Y_1(M)_{\mathbb{F}_p})^\times$, an $\mathbb{F}_p$-rational function, with $\overline{G}\circ\langle d\rangle = \chi_\ell(d)\overline{G}$ and $\overline{G}^{24} = \delta^*\overline{D}_\ell^{-1}$ (identities in R reduce). **What it still imports:** existence and smoothness of the μ_M-model with its $\mathbb{Z}[1/M]$-rational cusp (DR/KM/Katz/Gross), the identification of the complex points with $\Gamma_1(M)\backslash\mathfrak{H}$ compatible with the μ_M-structure (standard; recomputed in Lemma 7.3), representability of $\langle d\rangle$ and δ over $\mathbb{Z}[1/M]$. The note's "all cusps" formulation is *also* correct (divisor argument on the regular arithmetic surface with irreducible special fibre, where the expansion at ∞ alone shows the vertical multiplicity is 0), but it was not carried out; the monic-equation argument is shorter and is the one adopted.

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4. Supersingular lift and Frobenius

4.1 Deuring models without changing the point

Let $\tilde{x} = (E_1, \iota)$ be any lift. E_1 is isogenous to E (through δ), hence supersingular. Replacing (E_1, ι) by an isomorphic pair does not change the geometric point of the fine moduli scheme nor $\overline{G}(\tilde{x})$. **Claim:** every supersingular j has a model over $\mathbb{F}_{p^2}$ with $\pi_{p^2} = [-p]$. *Proof recorded in Lemma 7.8:* for any model over $\mathbb{F}_{p^2}$, $F : E \to E^{(p)}$ and $V : E^{(p)} \to E$ are both purely inseparable of degree p, so $F^{(p)} = u \circ V$ with $u \in \mathrm{Aut}(E)$ and $\pi_{p^2} = F^{(p)}F = u\,[p]$. For $j \neq 0, 1728$, $u = \pm 1$ and a quadratic twist flips the sign; for $j = 0$ ($p \equiv 2 \bmod 3$) and $j = 1728$ ($p \equiv 3 \bmod 4$) the curves $y^2 = x^3 + 1$, $y^2 = x^3 + x$

over $\mathbb{F}_p$ have trace 0, so $\pi_{p^2} = \pi_p^2 = [-p]$ (Deuring, 1941; Waterhouse, 1969; Ono, 2026, Lemma 4.3). The level structure ι is **not** required to be $\mathbb{F}_{p^2}$-rational, and in general it is not (it is defined over $\mathbb{F}_{p^{2r}}$, r the order of $-p^{-1}$ in $(\mathbb{Z}/M)^\times$). The "24-isogenous" phrase of the note is replaced by "supersingular".

4.2 Three actions distinguished

- **Frobenius on points.** $\sigma \in \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_{p^2})$, $u \mapsto u^{p^2}$, acts on $E_1(\overline{\mathbb{F}}_p)$ through the $\mathbb{F}_{p^2}$-structure, i.e. as the endomorphism $\pi_{E_1} = [-p]$; on $\mu_M(\overline{\mathbb{F}}_p)$ as $\zeta \mapsto \zeta^{p^2}$.
- **Galois action on the level structure.** $\iota^\sigma = \sigma_{E_1} \circ \iota \circ \sigma_{\mu_M}^{-1}$, so $\iota^\sigma(\zeta) = [-p]\iota(\zeta^{p^{-2}}) = [-p^{-1}]\iota(\zeta)$: $\sigma(\tilde{x}) = \langle -p^{-1}\rangle \tilde{x}$ in the μ_M-convention (fixed; the point convention would give $\langle -p\rangle'$).
- **Pull-back on functions.** $\overline{G}$ is $\mathbb{F}_p$-rational, so $\overline{G}(\sigma\tilde{x}) = \sigma(\overline{G}(\tilde{x})) = \overline{G}(\tilde{x})^{p^2}$; and $\overline{G} \circ \langle d\rangle = \chi_\ell(d)\overline{G}$.

Hence $\overline{G}(\tilde{x})^{p^2} = \chi_\ell(-p^{-1})\overline{G}(\tilde{x}) = \chi_\ell(p)\overline{G}(\tilde{x})$, using only *afterwards* that χ_ℓ is even and quadratic. Because $\overline{G}$ is a unit, $\overline{G}(\tilde{x})^{p^2-1} = \left(\frac{\ell}{p}\right)$ — **with no coefficient-field factor**: the only Galois action on coefficients is trivial because $\overline{G}$ is defined over $\mathbb{F}_p$ (and would still be trivial for σ if $\overline{G}$ were only defined over $\mathbb{F}_{p^2}$, as in the point model).

4.3 Passage to $D_\ell(x)^k$

$\overline{G}(\tilde{x})^{24} = \overline{D}_\ell(\delta\tilde{x})^{-1} = D_\ell(x)^{-1}$, so $D_\ell(x)^k = \overline{G}(\tilde{x})^{-24k} = \overline{G}(\tilde{x})^{-(p^2-1)} = \left(\frac{\ell}{p}\right)$. For $\ell = N \in \{2,3,5,7,13\}$, $D_N = u_N^{N-1}$ and $(N-1)k = (p^2-1)/a_N$.

$\ell = 2, 3$, small p, $j = 0, 1728$. The argument is uniform: $M = 576\ell$ is prime to $p \geq 5$ for every $\ell \neq p$; the fine moduli scheme $Y_1(M)$ has no stacky points, so $j = 0, 1728$ cause no difficulty; $D_\ell(x)$ is defined by the moduli formula regardless of $\mathrm{Aut}(E, C)$. Computationally (§8) the identity was verified at **every** geometric supersingular point of $X_0(\ell)$ for all $\ell \leq 47$, $5 \leq p \leq 199$, $p \neq \ell$, including all points over $j = 0$ and $j = 1728$ and the cases $\ell = 2, 3$ with $p = 5, 7$.

A consistency remark that the lemma predicts and Ono's Remark 4 observes: $\overline{G}(\tilde{x}) \in \mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$ exactly when $\left(\frac{\ell}{p}\right) = -1$ (e.g. $p = 11, \ell = 2$: "generate $\mathbb{F}_{11^4}$").

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5. The immediate corollary (constancy of $c(E)$)

$c(E) = B(E,\omega)^{(p-1)/2}/\Delta(E,\omega)^{(p^2-1)/24}$ has weight $\frac{(p+1)(p-1)}{2} - 12 \cdot \frac{p^2-1}{24} = 0$, so it is independent of ω and of the model — **invariance under isomorphism holds**. The inherited code evaluates it on $y^2 = 4x^3 - g_2x - g_3$ with $(E_4, E_6) = (12g_2, -216g_3)$, which is the correct normalisation for $\omega = dx/y$ (it matches $c_4 = 12g_2$, $c_6 = 216g_3$, $E_6 = -c_6$); any rescaling $(\lambda^4 E_4, \lambda^6 E_6)$ leaves c unchanged anyway.

Isogeny comparison. Lemma `Bisog`: $B(E', \omega') = \ell\, B(E, \omega)$ for $\omega = \pi^*\omega'$ at supersingular points. The note's q-expansion proof is sound ($E_2 - \ell E_2(\ell\tau)$ is a holomorphic weight-2 form on $\Gamma_0(\ell)$ with integral q-expansion; $B \equiv E_2 \bmod p$ by Kummer; q-expansion principle; $A = 0$ at x; $\ell^{p-1} = 1$). It is moreover a known result: Robert (1980, Théorème B/Lemme 7) as quoted in Baker (1998, Lemma 2.2): $B(E_2) = \lambda^{-(p+1)} \deg\phi\, B(E_1)$ for $\phi^*\omega_2 = \lambda\omega_1$ — our case is $\lambda = 1$. Verified numerically at every point of §8 (`robert_B_relation_ok`).

Graph connectivity. The ℓ-isogeny graph on supersingular j-invariants over $\overline{\mathbb{F}}_p$ is connected for every prime $\ell \neq p$ (Mestre, 1986; Pizer, 1990; Kohel, 1996) — standard.

Conclusion. With Theorem C for one prime ℓ: $c(E')/c(E) = \ell^{(p-1)/2}\,(\ell^{-12}D_\ell(x))^k = \left(\frac{\ell}{p}\right) \cdot 1 \cdot \left(\frac{\ell}{p}\right) = 1$ along every ℓ-isogeny, so c is constant by connectivity; $c^2 = \beta = -\left(\frac{-1}{p}\right)$ by Baker, hence $c \in \mathbb{F}_p$. **Corollary D is valid.** Conversely constancy of c for one ℓ implies Theorem C for that ℓ, and constancy for all ℓ is equivalent to Theorem C for all ℓ (given Lemma `Bisog`). **Alternative route:** none found. Ono's Theorem 1.3 controls only $\Delta^{(p^2-1)/12}$ (12th powers); the sign of the 24th-root question is exactly what his method cannot see (his Remark 4). No independent level-one argument for the constancy of c was found, so Corollary D does not currently provide a second proof of Theorem C. (The conjectural evaluation $c_p = \left(\frac{6}{p}\right)\left(\frac{p-1}{2}\right)!$ was not investigated, as instructed; no literature statement of it was encountered.)

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6. Literature and priority audit

Source (version inspected)	Where	Exact statement / normalisation	Relation to our identity
Ono, K. *Modular forms for chromatic homotopy: Supersingular congruences*, arXiv:2509.16175 **v2, 3 July 2026** (arXiv abs page lists v2 as current; PDF header "arXiv:2509.16175v2 [math.AT] 3 Jul 2026"; v1 (19 Sep 2025) has none of the items below).	Thm 1.2	$R_\ell(\tilde{P})^m = 1$, $m = (p^2-1)/12$, $R_\ell = V_\ell(\Delta)/\Delta = \Delta(\ell\tau)/\Delta(\tau) = D_\ell^{-1}$, every supersingular point of $X_0(\ell)$, $p \geq 5$, $\ell \neq p$.	The square of our identity.
	Thm 1.3	$\kappa(E) = \Delta(E,\omega)^{(p^2-1)/12} = (-1)^{(p+1)/2}$ for $E/\mathbb{F}_{p^2}$ supersingular with $\pi_E = [-p]$, ω $\mathbb{F}_{p^2}$-rational.	Different object (weight p^2-1 on Deuring models); **verified numerically here** (`deuring_kappa_ok`). With Baker it gives $B(E,\omega)^{p-1} = 1$ on Deuring models (also verified).
	§4.1–4.2, Lemma 4.3, Lemma 4.4	Vélu-normalised $\Delta^{[\ell]}(E,C,\omega) = \ell^{-12}\Delta(E',\omega')$; $R_\ell(\tilde{P})^m = \kappa(E')/\kappa(E)$; Deuring models.	Same moduli normalisation as ours (so $R_\ell = D_\ell^{-1}$ exactly).
	Remark 4	$(p^2-1)/24$ power can be -1: $p = 11, \ell = 2$, $R_2 \in \{6,7,2\}$; values of $g_\ell = \eta(\ell z)/\eta(z)$ need not lie in $\mathbb{F}_{p^2}$; proofs descending $\Delta^{1/24}$ or $\Delta^{1/12}$ to $\mathbb{F}_{p^2}$ must fail.	**Does not state our identity, nor any formula for the sign.** Our result requires a substantive additional argument (diamond characters on $X_1(576\ell)$), and is consistent with every example there.
Baker, A. (1998). *Acta Arith.* 86(1), 91–100 (author's copy read).	Thm 1.4	$B^{p-1} \equiv -\left(\frac{-1}{p}\right)\Delta^{(p^2-1)/12} \pmod{(p,A)}$, $B = E_{p+1}$, $A = E_{p-1}$.	Imported as Fact I3; **verified numerically** at every point of §8. No $(p^2-1)/24$ or $B^{(p-1)/2}$ statement.
	§2	For E_1, E_2 separably isogenous over $\mathbb{F}_{p^2}$: $\left(\left(\frac{-1}{p}\right)\Delta(E_i)\right)^{(p^2-1)/12}$ coincide (from Gross–Landweber).	Isogeny-invariance of Ono's κ over $\mathbb{F}_{p^2}$; with Ono's Lemma 4.4 it gives his Thm 1.2. Square of ours.
	Lemma 2.2, Cor. 2.3 (= Robert 1980, Thm B / Lemme 7)	$B(E_2) = \lambda^{-(p+1)} \deg\phi\, B(E_1)$ if $\phi^*\omega_2 = \lambda\omega_1$; $B(E_2)^{p-1} = \lambda^{-(p^2-1)} B(E_1)^{p-1}$.	**Precedent for the note's Lemma `Bisog`** (which should cite it). Does not involve Δ-ratios.
	(0.1) "Gross–Landweber congruence"	$u_2 \equiv \left(\frac{-1}{p}\right)\Delta^{(p^2-1)/12} \bmod (p, u_1)$, attributed to Gross & Landweber in the text, cited as Landweber (1988, LNM 1326, 69–93).	Source of the $(p^2-1)/12$ statements; not the $/24$ one.

Source (version inspected)	Where	Exact statement / normalisation	Relation to our identity
Robert, G. (1980). *Invent. Math.* 61, 103–158.	Thm B / Lemme 7	Known only through Baker's quotation.	**Full text not accessible in this session** (GDZ link not fetchable). Must be inspected for any half-exponent statement; flagged.
Kaneko & Zagier (1998), §10, full text read.	(35)–(40)	(39) de Shalit's signed residue formula at level 1; (40) $ss'_p(j)^{p-1} = \ldots$ (proved by Baker).	Exponents $p \pm 1$ only; no $/24$, no eta quotients, no level N.
de Shalit (1994)	Thm 1, Cor. 3	level-1 signed formula via p-adic periods.	Not about D_ℓ.
Landweber (1988), LNM 1326, 69–93	—	Congruences for Legendre polynomials / u_2	Not inspected directly (title/pages from Baker's bibliography).
Behrens (2009), *Geom. Topol.* 13, 319–357; Larson (2020), *Topology Appl.* 273, 107094	—	Divided β family; Larson's conjecture (proved by Ono v2 Thm 1.1).	Not inspected beyond Ono's bibliography; the brief's "Behrens and Larson" appears to mean these two separate works.
Gordon & Hughes (1993); Newman (1959); Ono (2004) Thm 1.64; Ligozat (1975)	—	Eta-quotient transformation law with character $\left(\frac{(-1)^k s}{d}\right)$.	Citation correct as used (§2.1); replaced by a complete derivation.
Knopp (1970) Ch. 4 §4.1; Apostol (1990) Thm 3.4; Rademacher (1931)	—	Closed form / Dedekind-sum form of the eta multiplier.	Import for §2.2; cross-checked exactly.
Katz (1976) §1; Gross (1990) §2 (read); Deligne–Rapoport (1973); Katz–Mazur (1985); Conrad (n.d.) handout (read); Loeffler (n.d.) §3.5 (read)	—	μ_M-model, rational cusp, q-expansion principle; point-model caveat.	Imports for §3.
Deuring (1941); Waterhouse (1969)	—	Deuring models / Frobenius $\pi = pu$.	Import for §4.1 (with an elementary proof recorded).
Mestre (1986); Pizer (1990); Kohel (1996)	—	Connectivity of the supersingular ℓ-isogeny graph.	Import for §5.
Betina–Lecouturier (TAMS 2019 final text); Robert (1980) full text; Landweber (1988); Behrens (2009); Larson (2020); Sakai (2011)	—	—	**Not inspected / inaccessible in this session.**

Separation of correctness from priority. Correctness of Theorem C and Corollary D is established above modulo the listed imports. On priority: the exact signed identity (and equivalently the constancy of $c(E)$) was not found in Ono v2, Baker, Kaneko–Zagier, de Shalit, or the eta-character literature; Ono v2 explicitly stops short of it. The closest statements are the $(p^2-1)/12$ results (Gross–Landweber → Baker §2 → Ono Thm 1.2). This is **not** a proof of novelty: Robert (1980) and the homotopy-theoretic literature (Behrens, Larson, Landweber) were not read in full.

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7. Dependency table

Step	Statement	Status	Source
1	G holomorphic, non-vanishing on $\mathfrak{H}$, q-expansion in $q^{\ell-1}\mathbb{Z}[[q]]$	proved (trivial)	—
2	Petersson/Knopp closed form of ν_η	**import**	Knopp 1970 Ch.4 §4.1; Apostol 1990 Thm 3.4; verified exactly vs Dedekind sums (400 matrices) and numerics
3	$G(\gamma\tau) = \left(\frac{\ell}{d}\right) G(\tau)$ on $\Gamma_0(576\ell)$	proved (§2.2)	from 2; verified exactly for $\ell \leq 47$, all classes of d
4	χ_ℓ = Kronecker character of $\mathbb{Q}(\sqrt{\ell})$, even, quadratic, $\chi_\ell(p) = \left(\frac{\ell}{p}\right)$	proved (reciprocity)	verified exhaustively mod 576ℓ
5	μ_M-model of $Y_1(M)$, $X_1(M)$: fine, smooth over $\mathbb{Z}[1/M]$, cusp ∞ a $\mathbb{Z}[1/M]$-section with parameter q, $\langle d\rangle$ and δ defined over $\mathbb{Z}[1/M]$, special fibre irreducible	**import**	DR 1973; KM 1985; Katz 1976 §1; Gross 1990 §2
6	$x_{\gamma\tau} = \langle d\rangle x_\tau$; $f \circ \langle d\rangle = \chi(d) f$	proved (lattice computation)	—
7	q-expansion principle ⇒ $G \in \mathbb{Q}(X_1(M))$	proved (2 lines) from 5	—
8	D_ℓ moduli-theoretic unit on $Y_0(\ell)_{\mathbb{Z}[1/\ell]}$ with expansion $\Delta(q)/\Delta(q^\ell)$	proved (Tate curve)	—
9	δ is $\tau \mapsto 24\tau$; $G^{24} = \delta^* D_\ell^{-1}$; δ surjective on $\overline{\mathbb{F}}_p$-points	proved	—
10	$G \in \mathcal{O}(Y_1(M)_{\mathbb{Z}_{(p)}})^\times$	proved (normality + monic equation) from 5,7,8,9	—
11	Deuring model for every supersingular j; subgroups rational; quotients keep $\pi = [-p]$	proved (Lemma 7.8) modulo existence of trace-0 curves at $j = 0, 1728$	Deuring 1941; Waterhouse 1969
12	$\sigma(\tilde{x}) = \langle -p^{-1}\rangle \tilde{x}$	proved	fine moduli (5)
13	$\overline{G}(\tilde{x})^{p^2-1} = \left(\frac{\ell}{p}\right)$	proved from 3,4,6,10,12	—
14	$D_\ell(x)^k = \left(\frac{\ell}{p}\right)$	proved from 9,13	—
15	Lemma `Bisog`: $B(E') = \ell B(E)$, $\Delta(\ell\tau)(x) = \ell^{-12}\Delta(E', \omega')$	proved in note; also **known**	Robert 1980 Lemme 7 via Baker 1998 Lemma 2.2
16	Baker's congruence $c^2 = -\left(\frac{-1}{p}\right)$	**import**	Baker 1998 Thm 1.4 (verified numerically)
17	connectivity of ℓ-isogeny graphs	**import**	Mestre 1986; Pizer 1990; Kohel 1996
18	Corollary D	proved from 14,15,16,17	—

8. Targeted reproducible checks

`code/velu_sign_check.py` (python-flint 0.9.0) computes, **without eta functions or Hauptmoduln**, for every supersingular $j \in \mathbb{F}_{p^2}$, a model with scalar p^2-Frobenius, all $\ell+1$ kernel polynomials from the ℓ-division polynomial, Vélu's normalised quotient E', and $r = \Delta(E)/\Delta(E') = \ell^{-12} D_\ell(E, C)$. Checks per point: $r^{(p^2-1)/12} = 1$ (Ono), $r^{(p^2-1)/24} = \left(\frac{\ell}{p}\right)$ (Theorem C), $B(E') = \ell B(E)$ (Robert/Baker, with $B = E_{p+1}$ mod p recomputed from scratch by linear algebra on q-expansions), Baker's congruence, $c(E') = c(E)$, $j(E')$ supersingular, $A(E) = 0$; per (ℓ, p): c constant over all supersingular j; for $N \in \{2, 3, 5, 7, 13\}$: the multiset $\{D_N(E, C)\}$ equals $\{u^{N-1} : P_N(u) = ju^N\}$, which ties the Vélu normalisation to the note's $u_N^{N-1} = \Delta/\Delta(N\tau)$; on Deuring models: Ono's $\kappa(E) = (-1)^{(p+1)/2}$ and $B(E, \omega) \in \mathbb{F}_p$.

Sweep: levels $\ell \in$ {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47}, primes $5 \le p \le 199$, $p \neq \ell$: 647 pairs, **127541 geometric supersingular points (E, C)** (7709 over j = 0, 7701 over j = 1728). Theorem C failures: none. Wall time 3804.4 s.

ℓ	pairs	p range	points	points j=0	points j=1728	D_ℓ^k $\in \{\pm 1\}$ and = (ℓ/p)	Ono $(p^2-1)/12$ power = 1	B(E′)=ℓB(E)	B∈F_p (Baker)	c constant	Hauptmodul match	κ (Ono 1.3)	B∈F_p (Deuring)
2	44	5–199	1125	69	69	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓
3	44	5–199	1500	92	92	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓
5	43	7–199	2244	132	138	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓
7	43	5–199	2992	184	176	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓
11	43	5–199	4476	264	264	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
13	43	5–199	5236	322	322	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓	all ✓
17	43	5–199	6714	396	414	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
19	43	5–199	7460	460	440	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
23	43	5–199	8928	528	528	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
29	43	5–199	11160	660	690	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
31	43	5–199	11904	736	704	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
37	43	5–199	14136	874	874	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
41	43	5–199	15582	924	966	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
43	43	5–199	16324	1012	968	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓
47	43	5–199	17760	1056	1056	all ✓	all ✓	all ✓	all ✓	all ✓	n/a	all ✓	all ✓

Pairs with $(\ell/p) = -1$ (so D_ℓ^k = −1 at every point): 346 of 647; with $(\ell/p) = +1$: 301. In every pair the set of values of D_ℓ^k over all points is a single element, equal to (ℓ/p).

Small cases ($\ell \in \{2,3\}$, $p \in \{5,7,11,13\}$):

ℓ	p	points	D_ℓ^k values	(ℓ/p)	j=0 pts	j=1728 pts
2	5	3	['4']	-1	3	0

ℓ	p	points	D_ℓ^k values	(ℓ/p)	j=0 pts	j=1728 pts
2	7	3	['1']	+1	0	3
2	11	6	['10']	-1	3	3
2	13	3	['12']	-1	0	0
3	5	4	['4']	-1	4	0
3	7	4	['6']	-1	0	4
3	11	8	['1']	+1	4	4
3	13	4	['1']	+1	0	0

`code/eta_multiplier_audit.py`: see §2 (exact Dedekind sums vs closed form vs numerics; character of G for $\ell \le 47$ on random $\gamma \in \Gamma_0(576\ell)$ covering all unit classes of d modulo the conductor, both signs; triviality on $\Gamma_1(576\ell)$; evenness/quadraticity/identification with the Kronecker character exhaustively mod 576ℓ; the note's Lemma `mult`; negative control).

Reproduce: `pip install python-flint==0.9.0 mpmath`; `python3 code/eta_multiplier_audit.py` (≈10 min, Dedekind sums are exact); `python3 code/velu_sign_check.py 31 2,3,5,7,11,13` (seconds) or `python3 code/velu_sign_check.py 199 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47` (≈1 h). Every check is a boolean in `results/*.json`.

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9. Residual points for a specialist

1. The imports in rows 2, 5, 11, 16, 17 of §7 are standard but were cited at section level (Knopp §4.1; DR IV–V; Gross §2 Prop. 2.1, verified from the text); the exact theorem numbers in Deligne–Rapoport and Katz–Mazur for the μ_M-model and its Tate-curve cusp were not checked line by line.
2. Robert (1980) should be read in full for (a) the precise statement of Lemme 7 (to cite it for Lemma `Bisog`), and (b) any statement involving $E_{p+1}^{(p-1)/2}$ or 24th roots of discriminant ratios.
3. The replacement section assumes the note's Lemma `Bisog` (unchanged) for the final sentence about $\Delta(\ell\tau)(x)$; the moduli definition (7.2) makes that sentence a definition rather than a lemma.
4. No new moonshine claims, no large-prime survey, and no investigation of Conjecture E were undertaken (as instructed).

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10. Files

```
AUDIT_REPORT.md / .pdf            this report
README.md                        layout, reproduction, verdict in one paragraph
replacement_section/
   eta_section_replacement.tex  drop-in replacement for Section 7 of the note (+ comments
    standalone_wrapper.tex/.pdf   compile test with the note's preamble
code/
    eta_multiplier_audit.py       §2 checks
    velu_sign_check.py            §8 checks
    make_tables.py                summary tables from the JSON
results/
   eta_multiplier_audit.json, velu_sign_checks.json, velu_sign_run.log, velu_sign_table.
literature/
   sources_inspected.md       every source, version/URL inspected, what was read
SHA256SUMS.txt
```

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